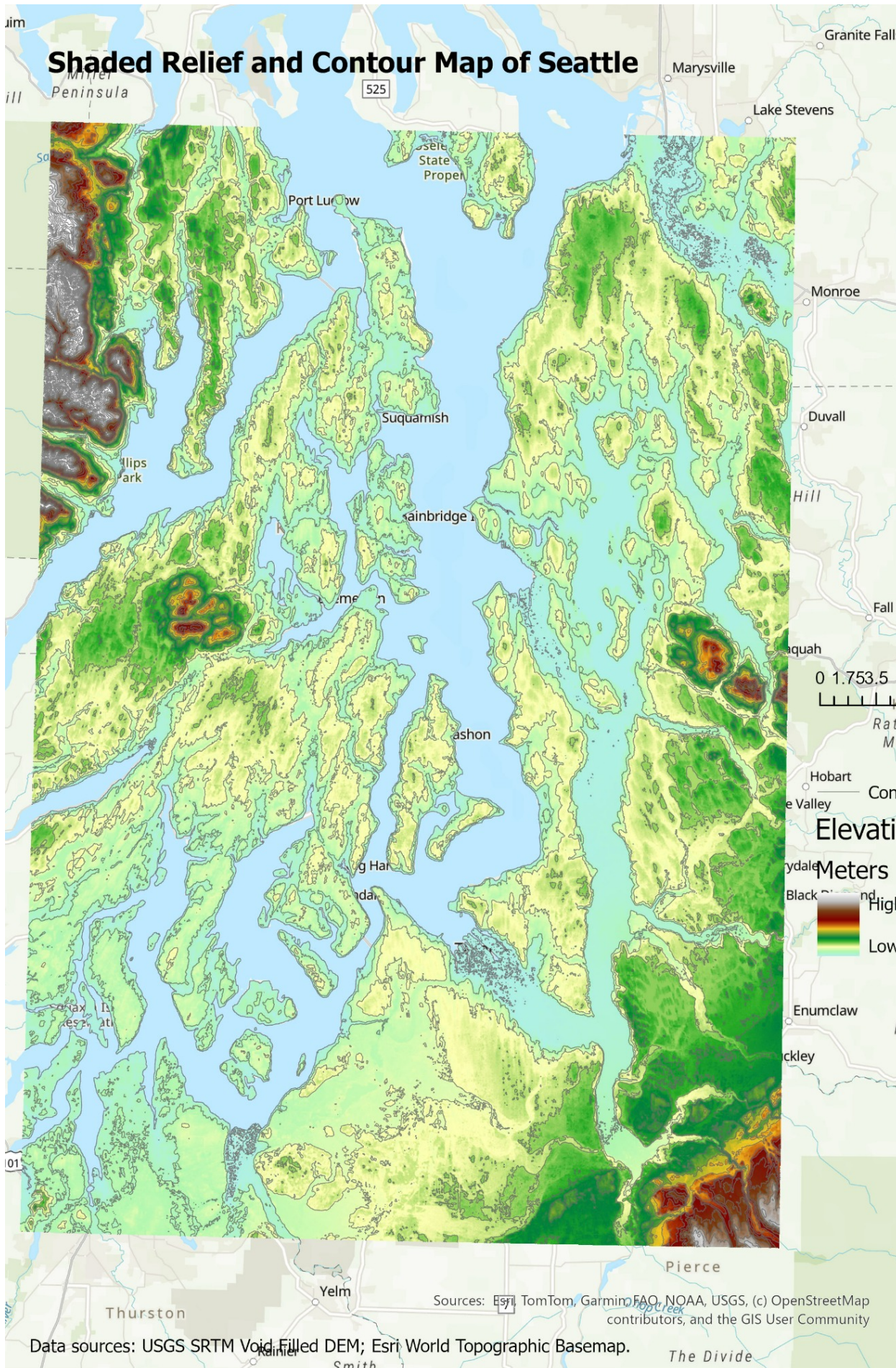


Shaded Relief and Contour Map of Seattle



Elevation (meters)



Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, (c) OpenStreetMap contributors, and the GIS User Community

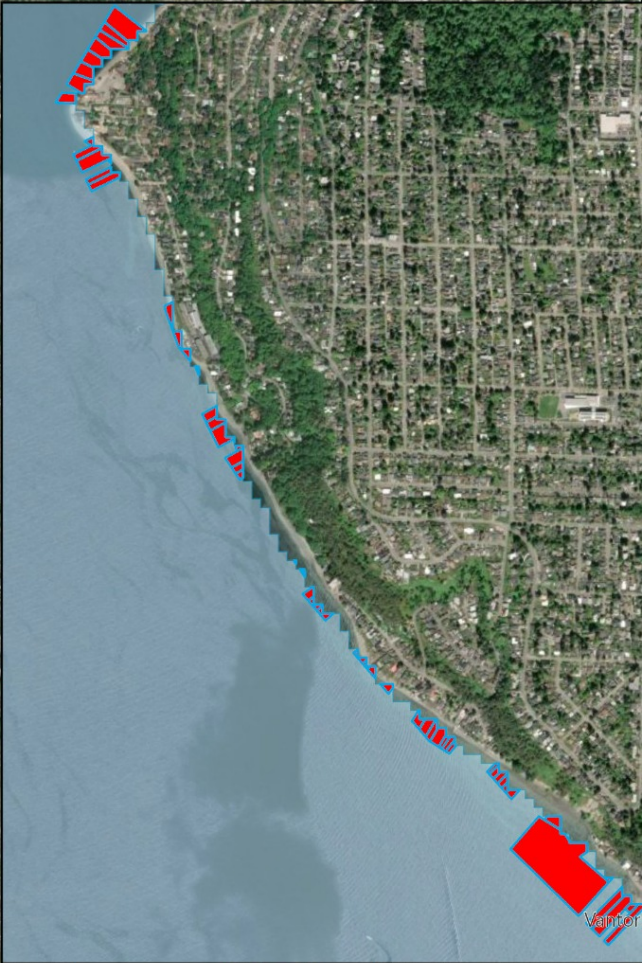
Data sources: USGS SRTM Void Filled DEM; Esri World Topographic Basemap.

Residential Zones Vulnerable to 1-Meter Sea Level Rise in Seattle by 2100

Data sources: USGS SRTM Void Filled DEM; Seattle GeoData Single and Multi Family Residential Zones; Esri World Imagery Basemap.



- Vulnerable Residential Zones
- Modeled 1-Meter Sea Level Rise



Written Report: Modeled 1-Meter Sea Level Rise in Seattle by 2100

This analysis models the potential impact of a 1-meter sea level rise scenario in Seattle by 2100. I selected 1 meter because it is a reasonable planning scenario within the projected range for Seattle. Seattle Public Utilities states that sea level rise in Seattle is projected to be approximately one foot by 2050 and between two and five feet by 2100. Since 1 meter is about 3.28 feet, this scenario falls within the city's projected 2100 range and represents a moderate-to-high risk planning estimate. NOAA's 2022 Sea Level Rise Technical Report also states that at least 2 feet, or about 0.6 meters, of sea level rise is likely along the U.S. coastline by 2100, with higher increases possible depending on future emissions.

To model this scenario, I used the USGS SRTM Void Filled DEM as the elevation dataset. In ArcGIS Pro, I used the Con tool to classify all raster cells with elevation values less than or equal to 1 meter as potentially inundated. Cells above 1 meter were excluded by converting them to NoData. I then converted the resulting inundation raster into a polygon layer and intersected it with the City of Seattle's Single and Multi Family Residential Zones dataset. This workflow allowed me to identify residential zones that spatially overlap with the modeled sea level rise zone.

The intersect analysis identified 288 Single and Multi Family Residential Zones as vulnerable to the modeled 1-meter sea level rise scenario. The total vulnerable residential land area is approximately 7,924,389 square feet, which equals about 0.736 square kilometers. The most visibly vulnerable areas appear to be concentrated along low-lying shoreline and waterway zones, especially near the Duwamish Waterway, South Park, Georgetown, and other waterfront-adjacent areas. These areas are more exposed because they sit close to tidal waters and contain relatively low-elevation land.

This map could support climate adaptation planning, emergency preparedness, zoning decisions, infrastructure investment, and environmental justice screening. For example, city planners could use it to identify where flood mitigation, drainage improvements, or long-term land-use changes may be most urgent. However, the map has several important limitations. The SRTM DEM is relatively coarse, so it may not capture small elevation differences at the parcel level. The model also does not account for storm surge, king tides, wave action, seawalls, drainage systems, or future infrastructure improvements. In addition, the 1-meter scenario is a simplified threshold model rather than a full hydrological flood simulation. Therefore, the results should be interpreted as an estimate of spatial exposure, not a precise prediction of future damage.

References

Seattle Public Utilities. Climate Change Projections & Maps.

<https://www.seattle.gov/utilities/protecting-our-environment/community-programs/climate-change/projections-and-maps>

U.S. Sea Level Change / NOAA et al. 2022 Sea Level Rise Technical Report. <https://earth.gov/sealevel/us/resources/2022-sea-level-rise-technical-report/>

USGS EarthExplorer. SRTM Void Filled DEM. <https://earthexplorer.usgs.gov/>

Seattle GeoData. Single and Multi Family Residential. <https://data-seattlecitygis.opendata.arcgis.com/datasets/SeattleCityGIS::single-and-multi-family-residential/>